

Bluetooth Controlled Car using Arduino

1. Objective

To design and implement a Bluetooth-controlled car using **Arduino Uno** and an **HC-05 Bluetooth module**, allowing wireless control of the car’s movement (forward, backward, left, right, and stop) through an Android smartphone.

2. Components Required

Component	Quantity	Description
Arduino Uno	1	Microcontroller board used for controlling the car
HC-05 Bluetooth Module	1	Enables wireless communication between Arduino and smartphone
L298N Motor Driver Module	1	Drives the DC motors based on signals from Arduino
DC Motors	4	Used for driving the wheels
Car Chassis	1	Mechanical body to mount all components
Wheels	4	For movement
9V or 12V Battery	1	Power supply for motors and Arduino
Jumper Wires	—	For electrical connections
Android Smartphone	1	Used to send control commands via Bluetooth app

3. Circuit Connections

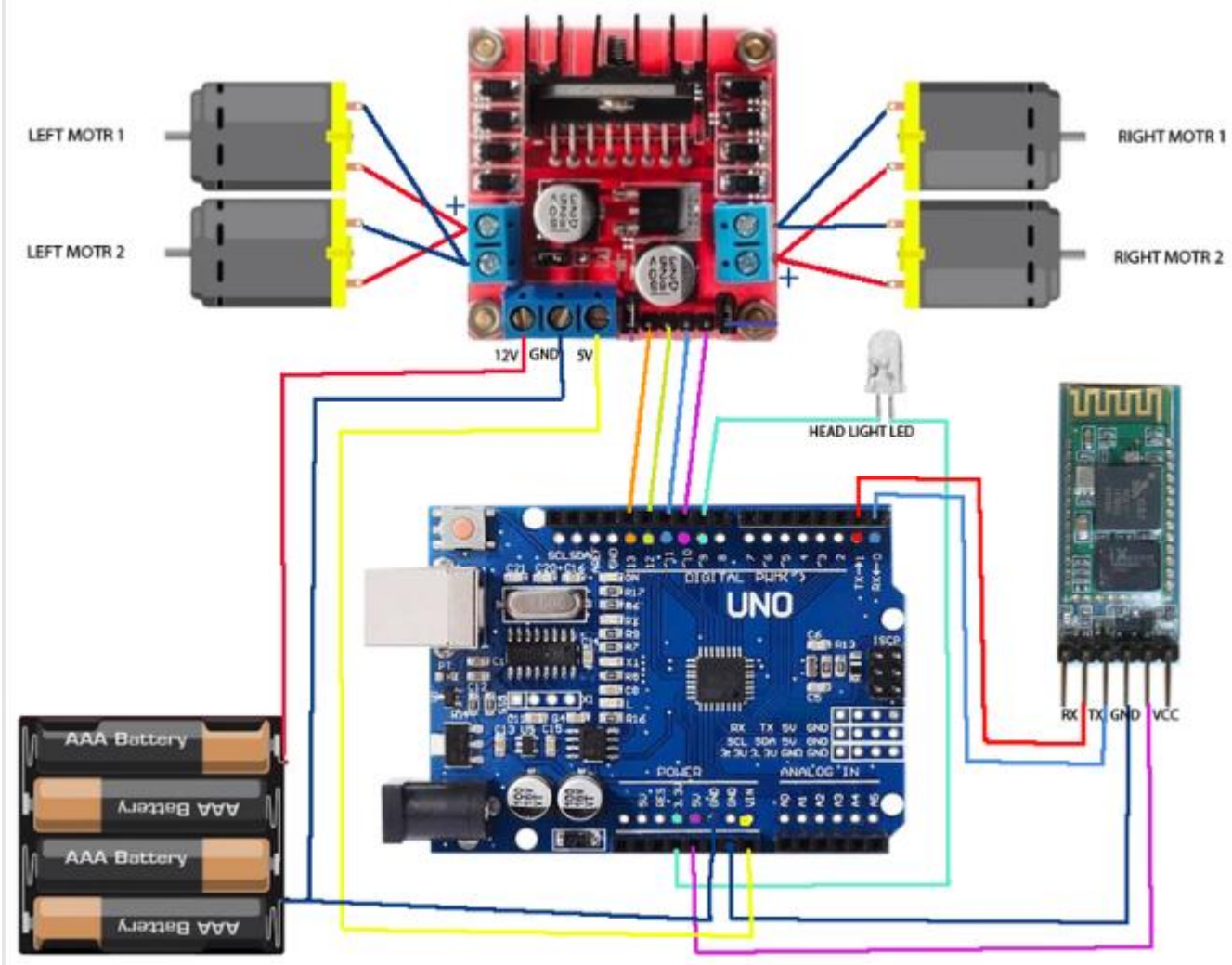
HC-05 Bluetooth Module

HC-05 Pin	Arduino Pin
VCC	5V
GND	GND
TXD	RX (Pin 0)
RXD	TX (Pin 1)

L298N Motor Driver Module

L298N Pin	Arduino Pin	Function
IN1	7	Controls Motor A direction
IN2	6	Controls Motor A direction
IN3	5	Controls Motor B direction
IN4	4	Controls Motor B direction
ENA	9 (PWM)	Controls Motor A speed
ENB	10 (PWM)	Controls Motor B speed
VCC	12V Battery +	
GND	Common Ground with Arduino	
5V	5V output (if jumper enabled)	

Important Note: Before uploading the code, **disconnect the TX and RX pins** of the HC-05, because they interfere with the USB connection. Reconnect them after the upload is complete.



4. Working Principle

The Bluetooth-controlled car works on the principle of **wireless communication** between a smartphone and the Arduino through the **HC-05 Bluetooth module**. The user controls the car from a smartphone app that sends some predefined characters whenever the corresponding button is pressed via Bluetooth which are received by the HC-05 module and transmitted to the Arduino's serial input pins (TX/RX).

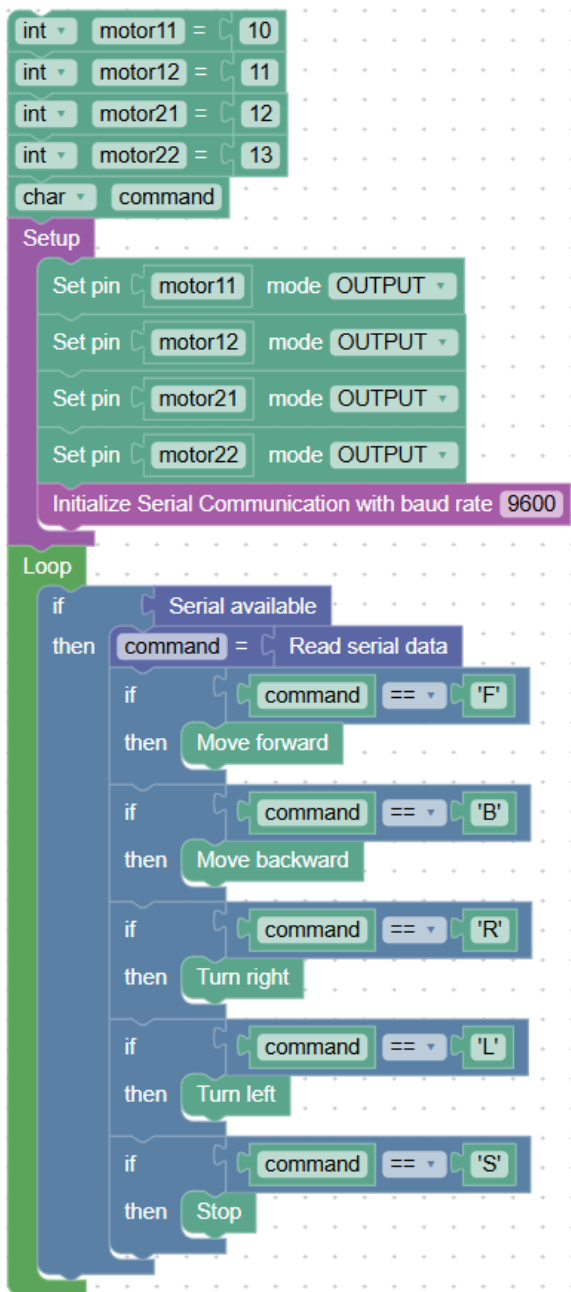
The Arduino reads these commands and decides the movement of the car by controlling the **L298N motor driver module**. The motor driver acts as an interface between the Arduino and the DC motors, allowing the motors to rotate forward or backward depending on the signals received. This allows full wireless control of the car's motion without any physical connection.

In summary, the **Bluetooth module** handles communication, the **Arduino** processes commands, and the **motor driver** executes the movement by controlling the motors.

5. Step by Step Method

1. Pair your smartphone with the **HC-05 Bluetooth module**.
2. Open the Bluetooth controller app and connect to **HC-05**.
3. When you press a direction button, the app sends a command (e.g., 'F').
4. Arduino receives this character via the serial port.
5. Depending on the command, the corresponding motor control pins are set HIGH or LOW.
6. The **L298N motor driver** activates the motors to move the car in the desired direction.

6. Block Based Code



7. Arduino Code

```
//define motor driver pins
int motor11 = 10;
int motor12 = 11;
int motor21 = 12;
int motor22 = 13;
char command; //variable to store incoming bluetooth commands
```

```
void setup() {
  //set pin type
  pinMode(motor11, OUTPUT);
  pinMode(motor12, OUTPUT);
  pinMode(motor21, OUTPUT);
  pinMode(motor22, OUTPUT);
}
```

```

Serial.begin(9600); //Start serial communication
}
void loop() {
  if (Serial.available()) {
    command = Serial.read(); //read command from bluetooth
    if (command == 'F') {
      //Turn all motors forward
      digitalWrite(motor11, HIGH);
      digitalWrite(motor12, LOW);
      digitalWrite(motor21, HIGH);
      digitalWrite(motor22, LOW);
    }
    if (command == 'B') {
      //Turn all motors backward
      digitalWrite(motor11, LOW);
      digitalWrite(motor12, HIGH);
      digitalWrite(motor21, LOW);
      digitalWrite(motor22, HIGH);
    }
    if (command == 'R') {
      //Turn left side motors forward and right backward
      digitalWrite(motor11, HIGH);
      digitalWrite(motor12, LOW);
      digitalWrite(motor21, LOW);
      digitalWrite(motor22, HIGH);
    }
    if (command == 'L') {
      //Turn right side motors forward and left backward
      digitalWrite(motor11, LOW);
      digitalWrite(motor12, HIGH);
      digitalWrite(motor21, HIGH);
      digitalWrite(motor22, LOW);
    }
    if (command == 'S') {
      //Stop all motors
      digitalWrite(motor11, LOW);
      digitalWrite(motor12, LOW);
      digitalWrite(motor21, LOW);
      digitalWrite(motor22, LOW);
    }
  }
}
}

```

8. Precautions

- Ensure a **common ground** between Arduino, L298N, and the Bluetooth module.
- Remove the **HC-05 module before uploading code**.
- Use a **sufficient power source** for motors (typically 9–12V).
- Avoid running both motors at full speed for long durations to prevent overheating.

9. Applications

- Educational robotics projects
- DIY hobby car projects
- Remote-controlled systems
- Foundation for AI or voice-controlled car systems